

Assignment#1: QA Tool

Subject: Software Quality Engineering



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SOFTWARE QUALITY TOOL: PLAYWRIGHT

1. Introduction:

Playwright is a current open source automation testing product created by Microsoft. It provides a means to do end to end testing of modern web applications. It is an automated web application tester, which clicks, types, explores, and validates browser interactions. Playwright allows programmers and testers to conduct end to end testing effectively using multiple browsers. Due to its first architecture of the browser, developers and QA engineers. Playwright is extremely fast and less susceptible to flakiness that web automation faces because it has direct access to the browsers internal protocol.

2. Key Features And Functionality:

- **Cross-browser:** Playwright is a rendering engine compatible with the existing rendering engines (Chromium, WebKit, and Firefox).
- **Cross language:** Support API in TypeScript, JavaScript, Python, .NET, Java.
- **Test Mobile Web:** Native mobile emulation of Android, Mobile Safari, and Google Chrome. The rendering engine is both Desktop and Cloud compatible.
- **Auto-wait:** Playwright will not take action if elements are actionable. Its introspection event set is also rich. The two together do away with artificial timeouts, which is the leading contributor to flaky tests.
- **Web-first assertions:** The web Playwright assertions are modeled to operate with the dynamic web. Checks will automatically be retried till the conditions needed are met.
- **Trace:** Set up test strategy, record execution trace, videos and screen shots to eliminate flakes..
- **Powerful Tooling:**
 - ★ **Codegen:** This is a command line tool used to record the interactions that a user can do in the browser and then automatically write the test code in the language of choice.
 - ★ **Trace Viewer:** This tool is a "time-travel" debugging tool that allows viewing of a complete recording of the given test, with the state of the DOM, network logs and console output per step.
 - ★ **Playwright inspector:** Inspect page, create selectors, debug the test execution, inspect click points, inspect execution logs.
- **Parallel Testing:** The test can be carried out in parallel to save time.

3. Other Tool Integrations:

Playwright can integrate easily with the DevOps ecosystem of today:

- **Programming Languages:** Playwright is compatible with various programming languages, such as TypeScript, JavaScript, Python, Java and C#/.NET.
- **CI/CD Pipelines:** It offers GitHub Actions, Azure Pipelines, Jenkins, GitLab and CircleCI official links and Docker images..
- **Test Runners:** Supports popular testing frameworks like Jest, Vitest, Mocha and Pytest.
- **Reporting:** HTML reporting, JSON, and JUnit are available, including third-party integrations, including Allure Report.

With these integrations, Playwright can be used in automated pipelines and collaboration among teams.

4. Cost And Licensing License:

- **License:** License: Playwright is published under the Apache License 2.0 and is therefore entirely Open Source and free to use, personally and commercially.
- **Cost:** Free of cost.

This renders it so user-friendly to students, startups, and businesses.

5. Advantages and Disadvantages:

PROS

- **Highperformance:** Tests have a lower response time than Selenium since it uses WebSockets.
- **Stability:** Stable auto-waiting is a major factor in minimizing flaky test results.
- **Multi-Tab/Frame Support:** Natively supports complex scenarios such as pop ups and iFrames.
- **Parallel Execution:** Does not require any additional infrastructure to be run in parallel.

CONS

- **Learning Curve:** Needs knowledge of programming (not a no code tool). Newer Community: It is a growing community, but less legacy tutorials than Selenium.
- **No Native Mobile:** Not able to test native ios/ android applications.
- **Modern Specs:** Modern Node.js or Python environments are required to use modern Specs.
- **Older versions:** Support of very old browsers is limited.

6. Industry Usage:

Playwright is also used in the software industry to:

- a. Web Application Testing.
- b. End-to-End Testing.
- c. UI Automation.
- d. Regression Testing.
- e. Constant Integration Pipelines.

It is commonly used by:

- i. QA Engineers.
- ii. Automation Testers.
- iii. Software Developers.
- iv. DevOps teams.

Several contemporary businesses are using Playwright because of its speed and reliability eg:

- **Automotive:** Mercedes-Benz makes use of Playwright to test their online dashboard and customer facing web sites.
- **E-commerce:** Retailers can simulate intricate checkout procedures with e-commerce and provide cross-browser functionality with Safari and Chrome clients.
- **Fintech:** Fintech is the term used by Alkami Technology and Shift4 to guarantee the trustworthiness and safety of banking interfaces.

7. Conclusion

Playwright is an effective and current software quality assurance product that makes the testing of web applications straightforward. Its cross browser support, automation, and integration functionality have earned it a formidable competitor to conventional tools. Though this involves a bit of knowledge in programming, it has very many benefits that make it very effective with beginner and experts in the software industry. It has established a new pace in speed and reliability. Playwright reflects the trade off between developer experience, cross browsing coverage and free of charge with no zerolicense, because any organization today wants to implement a sound Continuous Integration (CI) strategy.

Source: <https://playwright.dev/>

The content is mostly sourced using the official documentation of the platform particularly in terms of features and functionalities, with artificial intelligence used to only make grammatical amends, but the content of the content is completely provided by humans.